

Use the below code and implement our CICD stages.

- Source Code: <https://github.com/betawins/hiring-app.git>
 - Repository contains the application code for the hiring app.
- Create one Declarative pipeline job
- Create one Scripted pipeline job
- Create one multi stage pipeline job
- Create one parallel stage pipeline job.

Create one Declarative pipeline job

Implementation Steps

1. Log in to Jenkins.
2. From the Jenkins dashboard, click **New Item**.
3. Enter the job name:

Hiring-App-Declarative

4. Select:

Pipeline

5. Click **OK**.
6. Scroll down to the **Pipeline** section.
7. For **Definition**, select:

Pipeline script

8. Enter the following **Declarative Pipeline code**:

```
pipeline {
```

```
    agent any
```

```
    tools {
```

```
        maven 'Maven-3.9.4'
```

```
}
```

```
environment {
```

```
    APP_NAME = 'HiringApp'
```

```
    GIT_URL = 'https://github.com/betawins/hiring-app.git'
```

```
    GIT_BRANCH = 'main'
```

```
    SONAR_SERVER = 'SonarQube'
```

```
    SONAR_URL = 'http://<Private-IP>:9000'
```

```
    NEXUS_URL = '<Private-IP>:8081'
```

```
    NEXUS_REPOSITORY = 'maven_releases'
```

```
    NEXUS_CREDENTIALS = 'nexus-admin'
```

```
    TOMCAT_URL = 'http:// <Private-IP>:8080'
```

```
    TOMCAT_CREDENTIALS = 'tomcat-jenkins'
```

```
    TOMCAT_CONTEXT = 'hiring'
```

```
    SLACK_CHANNEL = 'all-jenkins'
```

```
    SLACK_CREDENTIALS = 'slack-jenkins-token'
```

```
    GROUP_ID = 'in.javahome'
```

```
    ARTIFACT_ID = 'hiring'
```

```
    APP_VERSION = '0.1'
```

```
    WAR_FILE = 'target/hiring.war'
```

```
}
```

```
stages {

    stage('Git Clone') {

        steps {

            deleteDir()

            git(
                branch: "${GIT_BRANCH}",
                url: "${GIT_URL}"
            )

            sh '''
                echo "=====
                echo "GIT CLONE"
                echo "=====

                git branch --show-current
                git log -1 --oneline

                echo ""
                echo "Repository files:"
                ls -la
            '''
        }
    }
}
```

```
stage('Maven Build') {

    steps {

        sh '''

            echo "=====
            echo "MAVEN BUILD"
            echo "=====

            mvn -version

            mvn clean package -DskipTests

            echo ""
            echo "Generated files:"
            ls -lh target/
            ""
        '''
    }
}
```

```
stage('SonarQube Analysis') {

    steps {

        withSonarQubeEnv("${SONAR_SERVER}") {

            sh ""
```

```

        echo "===== "
        echo "SONARQUBE ANALYSIS"
        echo "===== "

        mvn \
        org.sonarsource.scanner.maven:sonar-maven-plugin:sonar \
        -Dsonar.projectKey=hiring-app \
        -Dsonar.projectName=HiringApp \
        -Dsonar.sources=. \
        -Dsonar.host.url=$SONAR_URL
    ""
}
}
}

stage('Verify WAR') {

    steps {

        sh ""
        echo "===== "
        echo "VERIFY WAR"
        echo "===== "

        if [ ! -f "$WAR_FILE" ]; then
            echo "ERROR: hiring.war was not generated."
            ls -lah target/
            exit 1
        fi
    }
}

```

```
fi
```

```
echo "WAR found:"
```

```
ls -lh "$WAR_FILE"
```

```
echo ""
```

```
echo "WAR contents:"
```

```
unzip -l "$WAR_FILE"
```

```
""
```

```
}
```

```
}
```

```
stage('Nexus Upload') {
```

```
  steps {
```

```
    nexusArtifactUploader(
```

```
      nexusVersion: 'nexus3',
```

```
      protocol: 'http',
```

```
      nexusUrl: "${NEXUS_URL}",
```

```
      groupId: "${GROUP_ID}",
```

```
      version: "${APP_VERSION}",
```

```
      repository: "${NEXUS_REPOSITORY}",
```

```
      credentialsId: "${NEXUS_CREDENTIALS}",
```

```
      artifacts: [
```

```
        [
```

```
          artifactId: "${ARTIFACT_ID}",
```

```
          classifier: "",
```

```
        file: "${WAR_FILE}",
        type: 'war'
    ]
]
)
}
}
```

```
stage('Deploy Tomcat') {
```

```
    steps {
```

```
        deploy(
```

```
            adapters: [
```

```
                tomcat9(
```

```
                    credentialsId: "${TOMCAT_CREDENTIALS}",
```

```
                    url: "${TOMCAT_URL}",
```

```
                    path: "
```

```
                )
```

```
            ],
```

```
            contextPath: "${TOMCAT_CONTEXT}",
```

```
            war: "${WAR_FILE}"
```

```
        )
```

```
    }
```

```
}
```

```
stage('Slack Notification') {
```

```

steps {

    slackSend(
        channel: "${SLACK_CHANNEL}",
        color: 'good',
        tokenCredentialId: "${SLACK_CREDENTIALS}",
        botUser: true,
        message: ""
Hiring App CI/CD SUCCESS
Application : ${APP_NAME}
Build : #${BUILD_NUMBER}
Branch : ${GIT_BRANCH}
SonarQube:
${SONAR_URL}
Nexus:
http://${NEXUS_URL}
Tomcat:
${TOMCAT_URL}/${TOMCAT_CONTEXT}/
Jenkins:
${BUILD_URL}
""

    )
}
}

}

post {

```



```

failure {

    slackSend(
        channel: "${SLACK_CHANNEL}",
        color: 'danger',
        tokenCredentialId: "${SLACK_CREDENTIALS}",
        botUser: true,
        message: ""

Hiring App CI/CD FAILED
Application : ${APP_NAME}
Build : #${BUILD_NUMBER}
Branch : ${GIT_BRANCH}
Please check Jenkins console output.
${BUILD_URL}
""

    )
}

success {

    echo "Hiring application deployed successfully."

    echo "Application URL:"

    echo "${TOMCAT_URL}/${TOMCAT_CONTEXT}/"
}
}
}

```

9. Click **Apply**.
10. Click **Save**.
11. Click **Build Now**.
12. Open the generated **build number**.
13. Click **Console Output**.
14. Verify that the stages execute in the following order:

Git Clone

↓

Maven Build

↓

SonarQube Analysis

↓

Verify WAR

↓

Nexus Upload

↓

Deploy Tomcat

↓

Slack Notification

15. Verify the SonarQube analysis is successful and shows:

ANALYSIS SUCCESSFUL

BUILD SUCCESS

16. Verify that the WAR file is generated:

target/hiring.war

17. Verify that the WAR file is uploaded to the Nexus repository:

maven_releases

with:

Group ID : in.javahome

Artifact ID : hiring

Version : 0.1

18. Verify that the application is deployed to Tomcat with the context:

/hiring

19. After successful deployment, access:

http://<Public-IP>/hiring/

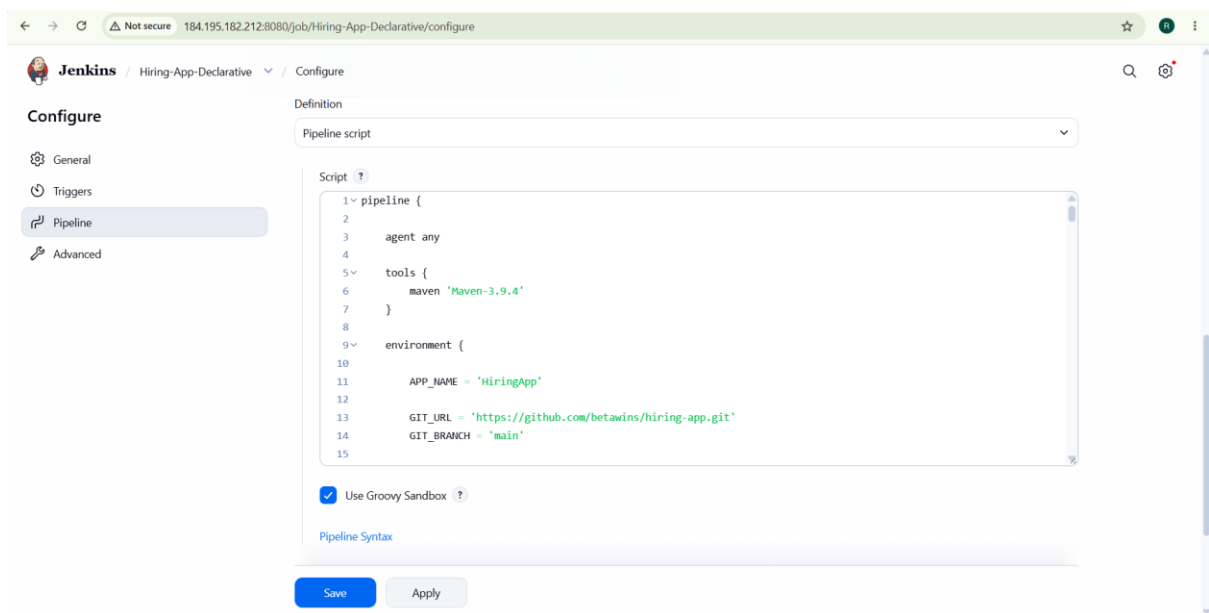
20. Verify that the **Hiring application** is accessible.

21. Verify that the **Slack success notification** is received in:

all-jenkins

22. Finally, verify that Jenkins shows:

Finished: SUCCESS



```
← → ↻ Not secure 184.195.182.212:8080/job/Hiring-App-Declarative/4/console ☆ ⓘ ⓘ
Jenkins / Hiring-App-Declarative / #4 🔍 ⓘ ⓘ

+ echo =====
+ echo 'GIT CLONE'
GIT CLONE
+ echo =====
+ git branch --show-current
main
+ git log -1 --oneline
9485b10 Update README.md
+ echo ''

+ echo 'Repository files:'
Repository files:
+ ls -la
total 44
drwxr-xr-x. 4 jenkins jenkins 171 Oct  8 09:33 .
drwxr-xr-x. 14 jenkins jenkins 16384 Oct  8 09:33 ..
drwxr-xr-x. 7 jenkins jenkins 146 Oct  8 09:33 .git
-rw-r--r--. 1 jenkins jenkins  8 Oct  8 09:33 .gitignore
-rw-r--r--. 1 jenkins jenkins 476 Oct  8 09:33 Dockerfile
-rw-r--r--. 1 jenkins jenkins 1757 Oct  8 09:33 Jenkinsfile
-rw-r--r--. 1 jenkins jenkins  88 Oct  8 09:33 README.md
-rw-r--r--. 1 jenkins jenkins 408 Oct  8 09:33 Untitled Diagram.drawio
-rw-r--r--. 1 jenkins jenkins 1881 Oct  8 09:33 jenkinsfile-cicd
-rw-r--r--. 1 jenkins jenkins 890 Oct  8 09:33 pom.xml
drwxr-xr-x. 3 jenkins jenkins 18 Oct  8 09:33 src
[Pipeline] }
[Pipeline] // withEnv
```

```
Jenkins / Hiring-App-Declarative / #4 🔍 ⓘ ⓘ

+ echo 'MAVEN BUILD'
MAVEN BUILD
+ echo =====
+ mvn -version
Apache Maven 3.9.4 (dfbb324ad4a7c8fb0bf182e6d91b0ae20e3d2dd9)
Maven home: /opt/maven
Java version: 21.0.12.1, vendor: Amazon.com Inc., runtime: /usr/lib/jvm/java-21-amazon-corretto.x86_64
Default locale: en, platform encoding: UTF-8
OS name: "linux", version: "6.18.51-120.163.amzn2023.x86_64", arch: "amd64", family: "unix"
+ mvn clean package -DskipTests
[INFO] Scanning for projects...
[INFO]
[INFO] -----< in.javahome:hiring >-----
[INFO] Building hiring Maven Webapp 0.1
[INFO] from pom.xml
[INFO] -----[ war ]-----
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ hiring ---
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ hiring ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /var/lib/jenkins/workspace/Hiring-App-Declarative/src/main/resources
[INFO]
[INFO] --- compiler:3.11.0:compile (default-compile) @ hiring ---
[INFO] No sources to compile
[INFO]
```

```
← → ↻ Not secure 184.195.182.212:8080/job/Hiring-App-Declarative/4/console ☆ ⓘ ⓘ
Jenkins / Hiring-App-Declarative / #4 🔍 ⓘ ⓘ

[INFO] No sources to compile
[INFO]
[INFO] --- surefire:3.1.2:test (default-test) @ hiring ---
[INFO] Tests are skipped.
[INFO]
[INFO] --- war:3.3.1:war (default-war) @ hiring ---
[INFO] Packaging webapp
[INFO] Assembling webapp [hiring] in [/var/lib/jenkins/workspace/Hiring-App-Declarative/target/hiring]
[INFO] Processing war project
[INFO] Copying webapp resources [/var/lib/jenkins/workspace/Hiring-App-Declarative/src/main/webapp]
[INFO] Building war: /var/lib/jenkins/workspace/Hiring-App-Declarative/target/hiring.war
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 1.249 s
[INFO] Finished at: 2026-10-08T09:33:58Z
[INFO] -----
+ echo ''

+ echo 'Generated files:'
Generated files:
+ ls -lh target/
total 4.0K
drwxr-xr-x. 4 jenkins jenkins 54 Oct  8 09:33 hiring
-rw-r--r--. 1 jenkins jenkins 1.7K Oct  8 09:33 hiring.war
drwxr-xr-x. 2 jenkins jenkins 28 Oct  8 09:33 maven-archiver
[Pipeline] }
[Pipeline] // withEnv
```

```
← → ↻ Not secure 184.195.182.212:8080/job/Hiring-App-Declarative/4/console ☆ ⓘ ⋮

Jenkins / Hiring-App-Declarative ▾ / #4 ▾ 🔍 ⚙️

[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (SonarQube Analysis)
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] withEnv
[Pipeline] {
[Pipeline] withSonarQubeEnv
Injecting SonarQube environment variables using the configuration: SonarQube
[Pipeline] {
[Pipeline] sh
+ echo =====
+ echo 'SONARQUBE ANALYSIS'
SONARQUBE ANALYSIS
+ echo =====
+ mvn org.sonarsource.scanner.maven:sonar-maven-plugin:sonar -Dsonar.projectKey=hiring-app -Dsonar.projectName=HiringApp -Dsonar.sources=. -
Dsonar.host.url=http://172.31.12.251:9000
[INFO] Scanning for projects...
Downloading from central: https://repo.maven.apache.org/maven2/org/sonarsource/scanner/maven/sonar-maven-plugin/maven-metadata.xml
Progress (1): 1.3 kB

Downloaded from central: https://repo.maven.apache.org/maven2/org/sonarsource/scanner/maven/sonar-maven-plugin/maven-metadata.xml (1.3 kB at
3.8 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/sonarsource/scanner/maven/sonar-maven-plugin/5.8.0.7211/sonar-maven-plugin-
5.8.0.7211.nv
```

```
← → ↻ Not secure 184.195.182.212:8080/job/Hiring-App-Declarative/4/console ☆ ⓘ ⋮

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[INFO] CPD Executor CPD calculation finished (done) | time=0ms
[INFO] SCM revision ID '9405b107fdffdf68e69fe953ea1f9669784f9d6a'
[INFO] Analysis report generated in 129ms, dir size=296.2 kB
[INFO] Analysis report compressed in 23ms, zip size=34.9 kB
[INFO] Analysis report uploaded in 40ms
[INFO] ANALYSIS SUCCESSFUL, you can find the results at: http://172.31.12.251:9000/dashboard?id=hiring-app
[INFO] Note that you will be able to access the updated dashboard once the server has processed the submitted analysis report
[INFO] More about the report processing at http://172.31.12.251:9000/api/ce/task?id=363310e8-88e9-4c23-8482-35df44a703d4
[INFO] Analysis total time: 4.136 s
[INFO] SonarScanner Engine completed successfully
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 10.083 s
[INFO] Finished at: 2026-10-08T09:34:10Z
[INFO] -----
[Pipeline] }
[Pipeline] // withSonarQubeEnv
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Verify WAR)
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] withEnv
[Pipeline] {
```

```
← → ↻ Not secure 184.195.182.212:8080/job/Hiring-App-Declarative/4/console ☆ ⓘ ⋮

Jenkins / Hiring-App-Declarative / #4 🔍 ⚙️

[Pipeline] { (Nexus Upload)
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] withEnv
[Pipeline] {
[Pipeline] nexusArtifactUploader
Uploading artifact hiring.war started....
GroupId: in.javahome
ArtifactId: hiring
Classifier:
Type: war
Version: 0.1
File: hiring.war
Repository:maven_releases
Uploading: http://172.31.47.243:8081/repository/maven_releases/in/javahome/hiring/0.1/hiring-0.1.war
100 % completed (1.7 kB / 1.7 kB).
Uploaded: http://172.31.47.243:8081/repository/maven_releases/in/javahome/hiring/0.1/hiring-0.1.war (1.7 kB at 5.3 kB/s)
Uploading artifact hiring.war completed.
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy Tomcat)
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] withEnv
[Pipeline] }
```

```
← → ↻ Not secure 184.195.182.212:8080/job/Hiring-App-Declarative/4/console ☆ ⓘ ⋮

Jenkins / Hiring-App-Declarative / #4 🔍 ⚙️

[Pipeline] withEnv
[Pipeline] {
[Pipeline] slackSend
Slack Send Pipeline step running, values are - baseUrl: <empty>, teamDomain: Jenkins, channel: all-jenkins, color: good, botUser: true,
tokenCredentialId: slack-jenkins-token, notifyCommitters: false, iconEmoji: <empty>, username: <empty>, timestamp: <empty>
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
Hiring application deployed successfully.
[Pipeline] echo
Application URL:
[Pipeline] echo
http://172.31.33.74:8080/hiring/
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

← → ↻ ⚠ Not secure 18.215.188.21:9000/dashboard?id=hiring-app&codeScope=new ☆ ⓘ ⋮

▲ Embedded database should be used for evaluation purposes only. It doesn't support scaling, upgrading to a new SonarQube Server version, or migration to another database engine. [Learn more](#) ↗

🏠 **SonarQube** community Projects Issues Rules Quality profiles Quality gates Administration More ▾ 🔍 🔔 ⓘ Ⓐ

HiringApp
Project

Analysis ▾
Overview
Issues
Security hot... **Deprecated**

Reporting >
Policies >
Project >
Project settings >

HiringApp / Overview

Overview

45 Lines of Code · Version 0.1 · Last analysis 18 minutes ago 🏠 Set as homepage ☆

✓ Quality gate ⓘ

Passed ↗

New Code

Overall Code

New Code Since not provided Started 25 minutes ago

New issues
0
Required = 0

Accepted issues
0
Valid issues that were not fixed

Coverage
A few extra steps are needed to set up coverage analysis.
[Set up coverage analysis](#) ↗

Duplications
-

Security Hotspots **Deprecated**
0

← → ↻ ⚠ Not secure 54.242.126.209:8081/#browse/browse:maven_releases ☆ ⓘ ⋮

sonatype nexus repository Community Edition 🔍 Search components or CVEs... 🔔 🔗 🔁 ⓘ Ⓐ

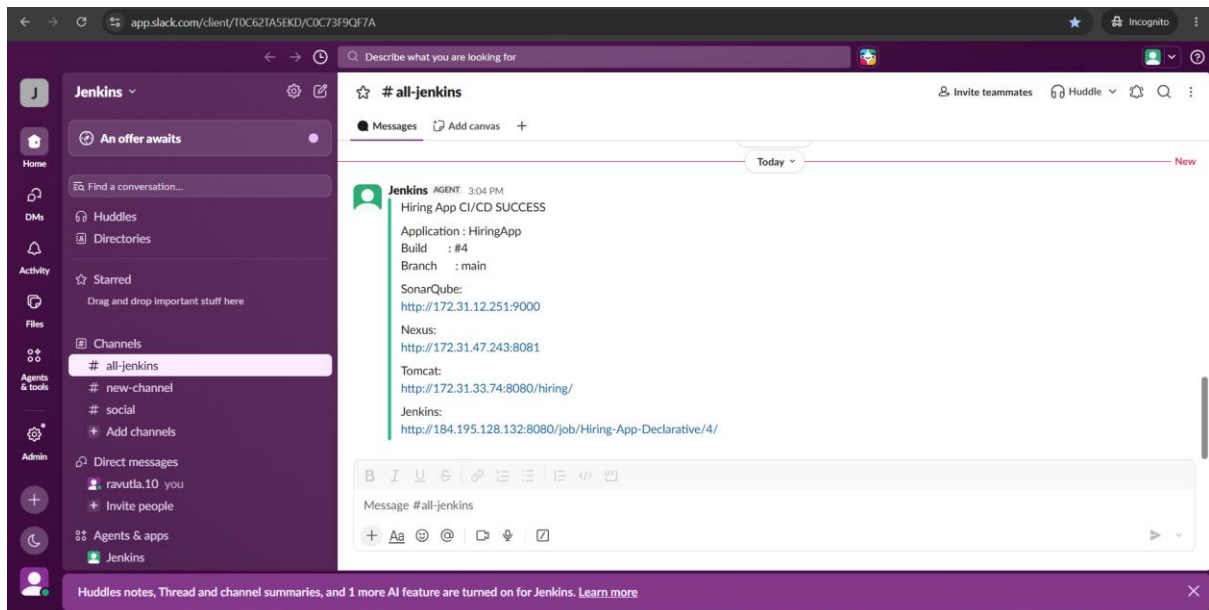
Browse / **maven_releases**

Upload component | [HTML View](#) [Advanced search...](#)

- com
 - javatpoint
 - SimpleCustomerApp
 - 1
 - 7
 - 8
 - 9
 - 16
 - in
 - javahome
 - hiring
 - 0.1
 - hiring-0.1.war
 - hiring-0.1.war.md5
 - hiring-0.1.war.sha1

← → ↻ ⚠ Not secure 54.157.35.48:8080/hiring/

Welcome to Techie Horizon



Create one Scripted pipeline job

Implementation Steps:

1. Login to **Jenkins**.
2. Go to **Jenkins Dashboard** → **New Item**.
3. Enter the job name:
Hiring-App-Scripted
4. Select **Pipeline**.
5. Click **OK**.
6. Scroll to the **Pipeline** section.
7. Under **Definition**, select:

Pipeline script

8. Enter the complete **Scripted Pipeline** code.

```
node {
```

```
def APP_NAME = 'HiringApp'
```



```
def GIT_URL = 'https://github.com/betawins/hiring-app.git'
```

```
def GIT_BRANCH = 'main'
```

```
def SONAR_SERVER = 'SonarQube'
```

```
def SONAR_URL = 'http:// <Private-IP>:9000'
```

```
def NEXUS_URL = '<Private-IP>:8081'
```

```
def NEXUS_REPOSITORY = 'maven_releases'
```

```
def NEXUS_CREDENTIALS = 'nexus-admin'
```

```
def TOMCAT_URL = 'http:// <Private-IP>:8080'
```

```
def TOMCAT_CREDENTIALS = 'tomcat-jenkins'
```

```
def TOMCAT_CONTEXT = 'hiring'
```

```
def SLACK_CHANNEL = 'all-jenkins'
```

```
def SLACK_CREDENTIALS = 'slack-jenkins-token'
```

```
def GROUP_ID = 'in.javahome'
```

```
def ARTIFACT_ID = 'hiring'
```

```
def APP_VERSION = '0.1'
```

```
def WAR_FILE = 'target/hiring.war'
```

```
stage('Git Clone') {
```

```
    deleteDir()
```

```
    git(
```

```

    branch: GIT_BRANCH,
    url: GIT_URL
)

sh '''
    echo "=====
    echo "GIT CLONE"
    echo "=====

    git branch --show-current
    git log -1 --oneline

    echo ""
    echo "Repository files:"
    ls -la
'''
}

```

```

stage('Maven Build') {

    def mavenHome = tool 'Maven-3.9.4'
    env.PATH = "${mavenHome}/bin:${env.PATH}"

    sh '''
        echo "=====
        echo "MAVEN BUILD"
        echo "=====
    '''
}

```

```
mvn -version
```

```
mvn clean package -DskipTests
```

```
echo ""
```

```
echo "Generated files:"
```

```
ls -lh target/
```

```
""
```

```
}
```

```
stage('SonarQube Analysis') {
```

```
withSonarQubeEnv(SONAR_SERVER) {
```

```
sh ""
```

```
echo "=====
```

```
echo "SONARQUBE ANALYSIS"
```

```
echo "=====
```

```
mvn \
```

```
org.sonarsource.scanner.maven:sonar-maven-plugin:sonar \
```

```
-Dsonar.projectKey=hiring-app \
```

```
-Dsonar.projectName=HiringApp \
```

```
-Dsonar.sources=. \
```

```
-Dsonar.host.url=$SONAR_HOST_URL
```

```
""
```

```
}  
}
```

```
stage('Verify WAR') {  
    sh """  
        echo "===== "  
        echo "VERIFY WAR"  
        echo "===== "  
  
        if [ ! -f "${WAR_FILE}" ]; then  
            echo "ERROR: hiring.war was not generated."  
            ls -lah target/  
            exit 1  
        fi  
  
        echo "WAR found:"  
        ls -lh "${WAR_FILE}"  
  
        echo ""  
        echo "WAR contents:"  
        unzip -l "${WAR_FILE}"  
    """"  
}
```

```
stage('Nexus Upload') {  
  
    nexusArtifactUploader(  

```

```
nexusVersion: 'nexus3',
protocol: 'http',
nexusUrl: NEXUS_URL,
groupId: GROUP_ID,
version: APP_VERSION,
repository: NEXUS_REPOSITORY,
credentialsId: NEXUS_CREDENTIALS,
artifacts: [
  [
    artifactId: ARTIFACT_ID,
    classifier: "",
    file: WAR_FILE,
    type: 'war'
  ]
]
)
```

```
stage('Deploy Tomcat') {
```

```
  deploy(
    adapters: [
      tomcat9(
        credentialsId: TOMCAT_CREDENTIALS,
        url: TOMCAT_URL,
        path: ""
      )
    ]
  )
}
```

```
    ],  
    contextPath: TOMCAT_CONTEXT,  
    war: WAR_FILE  
  )  
}
```

```
stage('Slack Notification') {
```

```
    slackSend(  
        channel: SLACK_CHANNEL,  
        color: 'good',  
        tokenCredentialId: SLACK_CREDENTIALS,  
        botUser: true,  
        message: ""
```

Hiring App CI/CD SUCCESS

Application : \${APP_NAME}

Build : #\${BUILD_NUMBER}

Branch : \${GIT_BRANCH}

SonarQube:

\${SONAR_URL}

Nexus:

http://\${NEXUS_URL}

Tomcat:

`${TOMCAT_URL}/${TOMCAT_CONTEXT}/`

Jenkins:

`${BUILD_URL}`

`""`

`)`

`}`

`}`

9. Click **Apply**.
10. Click **Save**.
11. Click **Build Now**.
12. Open the generated **Build Number**.
13. Click **Console Output**.
15. Verify **SonarQube Analysis** completes successfully.
16. Verify target/hiring.war is generated and verified.
17. Verify the WAR is uploaded successfully to **Nexus**.
18. Verify the application is deployed successfully to **Tomcat**.
19. Verify the **Slack success notification** is received.
20. Verify the final result:

Finished: SUCCESS

← → ↻ Not secure 184.195.182.212:8080/job/Hiring-App-Scripted/configure

Jenkins / Hiring-App-Scripted / Configure

Configure

- General
- Triggers
- Pipeline
- Advanced

Pipeline

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script

```
Script ?
1 node {
2
3   def APP_NAME = 'HiringApp'
4
5   def GIT_URL = 'https://github.com/betawins/hiring-app.git'
6   def GIT_BRANCH = 'main'
7
8   def SONAR_SERVER = 'SonarQube'
9   def SONAR_URL = 'http://172.31.12.251:9000'
10
11  def NEXUS_URL = '172.31.47.243:8081'
12  def NEXUS_REPOSITORY = 'maven_releases'
13  def NEXUS_CREDENTIALS = 'nexus-admin'
14
15  def TOMCAT_URL = 'http://172.31.33.74:8080'
```

Save Apply

← → ↻ Not secure 184.195.182.212:8080/job/Hiring-App-Scripted/3/console

Jenkins / Hiring-App-Scripted / #3

```
Commit message: "Update README.md"
> /usr/bin/git rev-list --no-walk 9405b107dfdf68e69f953eaf96e9784f9d6a # timeout=10
[Pipeline] sh
+ echo =====
+ echo 'GIT CLONE'
GIT CLONE
+ echo =====
+ git branch --show-current
main
+ git log -1 --oneline
9405b10 Update README.md
+ echo ''

+ echo 'Repository files:'
Repository files:
+ ls -la
total 44
drwxr-xr-x. 4 jenkins jenkins 171 Oct  8 10:06 .
drwxr-xr-x. 16 jenkins jenkins 16384 Oct  8 10:06 ..
drwxr-xr-x. 7 jenkins jenkins 146 Oct  8 10:06 .git
-rw-r--r--. 1 jenkins jenkins  8 Oct  8 10:06 .gitignore
-rw-r--r--. 1 jenkins jenkins 476 Oct  8 10:06 Dockerfile
-rw-r--r--. 1 jenkins jenkins 1757 Oct  8 10:06 Jenkinsfile
-rw-r--r--. 1 jenkins jenkins  88 Oct  8 10:06 README.md
-rw-r--r--. 1 jenkins jenkins 488 Oct  8 10:06 Untitled Diagram.drawio
-rw-r--r--. 1 jenkins jenkins 1801 Oct  8 10:06 jenkinsfile-cicd
```



```
← → ↻ ⚠ Not secure 184.195.182.212:8080/job/Hiring-App-Scripted/3/console ☆ ⓘ ⋮
Jenkins / Hiring-App-Scripted / #3 🔍 ⚙️
[Pipeline] cool
[Pipeline] sh
+ echo =====
+ echo 'MAVEN BUILD'
MAVEN BUILD
+ echo =====
+ mvn -version
Apache Maven 3.9.4 (dfbb324ad4a7c8fb0bf182e6d91b0ae20e3d2dd9)
Maven home: /opt/maven
Java version: 21.0.12.1, vendor: Amazon.com Inc., runtime: /usr/lib/jvm/java-21-amazon-corretto.x86_64
Default locale: en, platform encoding: UTF-8
OS name: "linux", version: "6.18.51-120.163.amzn2023.x86_64", arch: "amd64", family: "unix"
+ mvn clean package -DskipTests
[INFO] Scanning for projects...
[INFO]
[INFO] -----< in.javahome:hireng >-----
[INFO] Building hiring Maven Webapp 0.1
[INFO] from pom.xml
[INFO] -----[ war ]-----
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ hiring ---
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ hiring ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /var/lib/jenkins/workspace/Hiring-App-Scripted/src/main/resources
[INFO]
[INFO] --- compiler:3.11.0:compile (default-compile) @ hiring ---
```

```
← → ↻ ⚠ Not secure 184.195.182.212:8080/job/Hiring-App-Scripted/3/console ☆ ⓘ ⋮
Jenkins / Hiring-App-Scripted / #3 🔍 ⚙️
[INFO] Tests are skipped.
[INFO]
[INFO] --- war:3.3.1:war (default-war) @ hiring ---
[INFO] Packaging webapp
[INFO] Assembling webapp [hiring] in [/var/lib/jenkins/workspace/Hiring-App-Scripted/target/hiring]
[INFO] Processing war project
[INFO] Copying webapp resources [/var/lib/jenkins/workspace/Hiring-App-Scripted/src/main/webapp]
[INFO] Building war: /var/lib/jenkins/workspace/Hiring-App-Scripted/target/hiring.war
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 1.258 s
[INFO] Finished at: 2026-10-08T10:06:56Z
[INFO] -----
+ echo ''

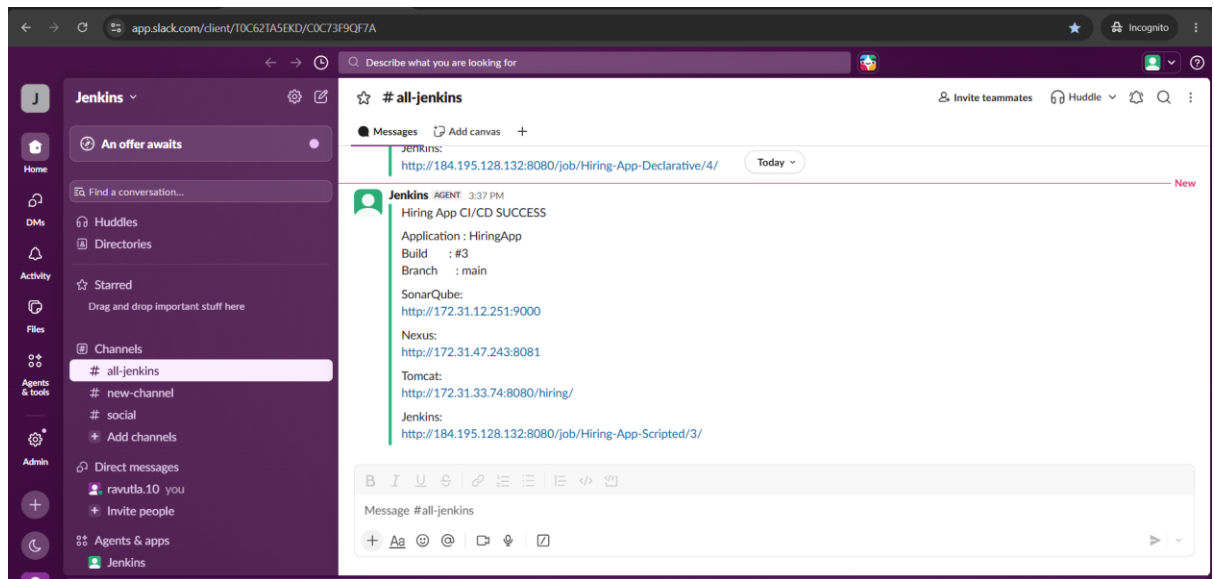
+ echo 'Generated files:'
Generated files:
+ ls -lh target/
total 4.0K
drwxr-xr-x. 4 jenkins jenkins 54 Oct 8 10:06 hiring
-rw-r--r--. 1 jenkins jenkins 1.7K Oct 8 10:06 hiring.war
drwxr-xr-x. 2 jenkins jenkins 28 Oct 8 10:06 maven-archiver
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (SonarQube Analysis)
[Pipeline] withSonarQubeEnv
Injecting SonarQube environment variables using the configuration: SonarQube
```

```
← → ↻ ⚠ Not secure 184.195.182.212:8080/job/Hiring-App-Scripted/3/console ☆ ⓘ
Jenkins / Hiring-App-Scripted / #3

[INFO] Sensor Zero Coverage Sensor
[INFO] Sensor Zero Coverage Sensor (done) | time=0ms
[INFO] ----- Gather SCA dependencies on project
[INFO] Dependency analysis skipped
[INFO] CPD Executor 1 file had no CPD blocks
[INFO] CPD Executor Calculating CPD for 0 files
[INFO] CPD Executor CPD calculation finished (done) | time=0ms
[INFO] SCM revision ID '9405b107fdffdf68e69fe953ealf9669784f9d6a'
[INFO] Analysis report generated in 142ms, dir size=296.2 kB
[INFO] Analysis report compressed in 24ms, zip size=34.9 kB
[INFO] Analysis report uploaded in 27ms
[INFO] ANALYSIS SUCCESSFUL, you can find the results at: http://172.31.12.251:9000/dashboard?id=hiring-app
[INFO] Note that you will be able to access the updated dashboard once the server has processed the submitted analysis report
[INFO] More about the report processing at http://172.31.12.251:9000/api/ce/task?id=368c5cc6-1dc6-451b-92c3-5daa0cfc4267
[INFO] Analysis total time: 3.961 s
[INFO] SonarScanner Engine completed successfully
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 7.948 s
[INFO] Finished at: 2026-10-08T10:07:06Z
[INFO] -----
[Pipeline] }
[Pipeline] // withSonarQubeEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Verify WAR)
[Pipeline] sh
fDinealnal sh
```

```
← → ↻ ⚠ Not secure 184.195.182.212:8080/job/Hiring-App-Scripted/3/console ☆ ⓘ
Jenkins / Hiring-App-Scripted / #3

[Pipeline] stage
[Pipeline] { (Nexus Upload)
[Pipeline] nexusArtifactUploader
Uploading artifact hiring.war started....
GroupId: in.javahome
ArtifactId: hiring
Classifier:
Type: war
Version: 0.1
File: hiring.war
Repository:maven_releases
Uploading: http://172.31.47.243:8081/repository/maven\_releases/in/javahome/hiring/0.1/hiring-0.1.war
100 % completed (1.7 kB / 1.7 kB).
Uploaded: http://172.31.47.243:8081/repository/maven\_releases/in/javahome/hiring/0.1/hiring-0.1.war (1.7 kB at 23 kB/s)
Uploading artifact hiring.war completed.
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy Tomcat)
[Pipeline] deploy
[DeployPublisher][INFO] Attempting to deploy 1 war file(s)
[DeployPublisher][INFO] Deploying /var/lib/jenkins/workspace/Hiring-App-Scripted/target/hiring.war to container Tomcat 9.x Remote with context hiring
  Redeploying [/var/lib/jenkins/workspace/Hiring-App-Scripted/target/hiring.war]
  Undeploying [/var/lib/jenkins/workspace/Hiring-App-Scripted/target/hiring.war]
  Deploying [/var/lib/jenkins/workspace/Hiring-App-Scripted/target/hiring.war]
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Slack Notification)
[Pipeline] slackSend
Slack Send Pipeline step running, values are - baseUrl: <empty>, teamDomain: Jenkins, channel: all-jenkins, color: good, botUser: true, tokenCredentialId: slack-jenkins-token, notifyCommiters: false, iconEmoji: <empty>, username: <empty>, timestamp: <empty>
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```



Create one multi stage pipeline job

Implementation Steps:

1. Hiring-App-Multi-Stage
2. Select **Pipeline**.
3. Click **OK**.
4. Scroll to the **Pipeline** section.
5. Under **Definition**, select:
Pipeline script
6. Enter the complete **Multi-Stage Pipeline** code.

```
pipeline {
```

```
  agent any
```

```
  tools {
```

```
    maven 'Maven-3.9.4'
```

```
  }
```

```
  environment {
```

```
    APP_NAME = 'HiringApp'
```

GIT_URL = 'https://github.com/betawins/hiring-app.git'

GIT_BRANCH = 'main'

SONAR_SERVER = 'SonarQube'

SONAR_URL = 'http:// <Private-IP>:9000'

NEXUS_URL = '<Private-IP>:8081'

NEXUS_REPOSITORY = 'maven_releases'

NEXUS_CREDENTIALS = 'nexus-admin'

TOMCAT_URL = 'http:// <Private-IP>:8080'

TOMCAT_CREDENTIALS = 'tomcat-jenkins'

TOMCAT_CONTEXT = 'hiring'

SLACK_CHANNEL = 'all-jenkins'

SLACK_CREDENTIALS = 'slack-jenkins-token'

GROUP_ID = 'in.javahome'

ARTIFACT_ID = 'hiring'

APP_VERSION = '0.1'

WAR_FILE = 'target/hiring.war'

}

stages {

stage('Source Code') {

steps {

deleteDir()

```
git branch: "${GIT_BRANCH}", url: "${GIT_URL}"
```

```
sh ""  
  
    echo "Git Clone Successful"  
  
    git log -1 --oneline  
  
    ""  
  
}  
  
}
```

```
stage('Build') {  
    steps {  
        sh ""  
  
        echo "Maven Build Started"  
  
        mvn -version  
  
        mvn clean package -DskipTests  
  
        echo "Maven Build Completed"  
  
        ls -lh target/hiring.war  
  
        ""  
  
    }  
  
}
```

```
stage('Code Quality') {  
    steps {  
        withSonarQubeEnv("${SONAR_SERVER}") {  
            sh ""  
  
            mvn \  
  
            org.sonarsource.scanner.maven:sonar-maven-plugin:sonar \  
  
            -Dsonar.projectKey=hiring-app \  

```

```

        -Dsonar.projectName=HiringApp \
        -Dsonar.sources=. \
        -Dsonar.host.url=$SONAR_HOST_URL
    ""
}
}
}

```

```

stage('Artifact') {
    steps {
        sh ""
        echo "Verifying WAR"
        test -f "${WAR_FILE}"
        ls -lh "${WAR_FILE}"
    ""
}

```

```

nexusArtifactUploader(
    nexusVersion: 'nexus3',
    protocol: 'http',
    nexusUrl: "${NEXUS_URL}",
    groupId: "${GROUP_ID}",
    version: "${APP_VERSION}",
    repository: "${NEXUS_REPOSITORY}",
    credentialsId: "${NEXUS_CREDENTIALS}",
    artifacts: [
        [
            artifactId: "${ARTIFACT_ID}",
            classifier: "",

```

```
        file: "${WAR_FILE}",
        type: 'war'
    ]
]
)
}
}
```

```
stage('Deployment') {
    steps {
        deploy(
            adapters: [
                tomcat9(
                    credentialsId: "${TOMCAT_CREDENTIALS}",
                    url: "${TOMCAT_URL}",
                    path: ""
                )
            ],
            contextPath: "${TOMCAT_CONTEXT}",
            war: "${WAR_FILE}"
        )
    }
}
```

```
stage('Notification') {
    steps {
        slackSend(
            channel: "${SLACK_CHANNEL}",
```

```
        color: 'good',
        tokenCredentialId: "${SLACK_CREDENTIALS}",
        botUser: true,
        message: ""
```

Hiring App CI/CD SUCCESS

Application : \${APP_NAME}

Build : #\${BUILD_NUMBER}

Branch : \${GIT_BRANCH}

Tomcat : \${TOMCAT_URL}/\${TOMCAT_CONTEXT}/

Jenkins : \${BUILD_URL}

""

```
    )
  }
}
}
```

```
post {
    failure {
        slackSend(
            channel: "${SLACK_CHANNEL}",
            color: 'danger',
            tokenCredentialId: "${SLACK_CREDENTIALS}",
            botUser: true,
            message: ""
```

Hiring App CI/CD FAILED

Application : \${APP_NAME}

Build : #\${BUILD_NUMBER}

Please check Jenkins console output.

`${BUILD_URL}`

`""`

`)`

`}`

`success {`

`echo "Hiring application deployed successfully."`

`echo "${TOMCAT_URL}/${TOMCAT_CONTEXT}"`

`}`

`}`

`}`

7. Click **Apply**.

8. Click **Save**.

9. Click **Build Now**.

10. Open the generated **Build Number**.

11. Click **Console Output**.

15. Verify **SonarQube Analysis** completes successfully.

16. Verify target/hiring.war is generated and uploaded to **Nexus**.

17. Verify the application is successfully deployed to **Tomcat**.

18. Verify the **Slack success notification** is received.

19. Verify the application:

`http://<Tomacat-Public IP>:8080/hiring/`

20. Verify the final result:

Finished: SUCCESS

← → ↻ Not secure 184.195.182.212:8080/job/Hiring-App-Multi-Stage/configure

Jenkins / Hiring-App-Multi-Stage / Configure

Definition

Configure

General

Triggers

Pipeline

Advanced

Pipeline script

```
1 pipeline {
2   agent any
3
4   tools {
5     maven 'Maven-3.9.4'
6   }
7
8   environment {
9     APP_NAME = 'HiringApp'
10
11     GIT_URL = 'https://github.com/betawins/hiring-app.git'
12     GIT_BRANCH = 'main'
13
14     SONAR_SERVER = 'SonarQube'
15     SONAR_URL = 'http://172.31.12.251:9000'
16 }
```

☒ Use Groovy Sandbox

[Pipeline Syntax](#)

Save Apply

← → ↻ Not secure 184.195.182.212:8080/job/Hiring-App-Multi-Stage/1/console

Jenkins / Hiring-App-Multi-Stage / #1

```
> /usr/bin/git init /var/lib/jenkins/workspace/Hiring-App-Multi-Stage # timeout=10
Fetching upstream changes from https://github.com/betawins/hiring-app.git
> /usr/bin/git --version # timeout=10
> git --version # 'git version 2.50.1'
> /usr/bin/git fetch --tags --force --progress -- https://github.com/betawins/hiring-app.git +refs/heads/*:refs/remotes/origin/* # timeout=10
> /usr/bin/git config remote.origin.url https://github.com/betawins/hiring-app.git # timeout=10
> /usr/bin/git config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
> /usr/bin/git rev-parse refs/remotes/origin/main^{commit} # timeout=10
Checking out Revision 9405b107fdffdf68e69fe953eaf9669784f9d6a (refs/remotes/origin/main)
> /usr/bin/git config core.sparsecheckout # timeout=10
> /usr/bin/git checkout -f 9405b107fdffdf68e69fe953eaf9669784f9d6a # timeout=10
> /usr/bin/git branch -a -v --no-abbrev # timeout=10
> /usr/bin/git checkout -b main 9405b107fdffdf68e69fe953eaf9669784f9d6a # timeout=10
Commit message: "Update README.md"
First time build. Skipping changelog.
[Pipeline] sh
+ echo 'Git Clone Successful'
Git Clone Successful
+ git log -1 --oneline
9405b10 Update README.md
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (build)
[Pipeline] tool
```

```
← → ↻ ⚠ Not secure 184.195.182.212:8080/job/Hiring-App-Multi-Stage/1/console ☆ ⓘ ⋮

Jenkins / Hiring-App-Multi-Stage / #1 🔍 ⚙️

[INFO] --- resources:3.3.1:testResources (default-testResources) @ hiring ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /var/lib/jenkins/workspace/Hiring-App-Multi-Stage/src/test/resources
[INFO] --- compiler:3.11.0:testCompile (default-testCompile) @ hiring ---
[INFO] No sources to compile
[INFO] --- surefire:3.1.2:test (default-test) @ hiring ---
[INFO] Tests are skipped.
[INFO] --- war:3.3.1:war (default-war) @ hiring ---
[INFO] Packaging webapp
[INFO] Assembling webapp [hiring] in [/var/lib/jenkins/workspace/Hiring-App-Multi-Stage/target/hiring]
[INFO] Processing war project
[INFO] Copying webapp resources [/var/lib/jenkins/workspace/Hiring-App-Multi-Stage/src/main/webapp]
[INFO] Building war: /var/lib/jenkins/workspace/Hiring-App-Multi-Stage/target/hiring.war
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 1.242 s
[INFO] Finished at: 2026-10-08T10:14:32Z
[INFO] -----
+ echo 'Maven Build Completed'
Maven Build Completed
+ ls -lh target/hiring.war
-rw-r--r-- 1 jenkins jenkins 1.7K Oct 8 10:14 target/hiring.war
[Pipeline] }
```

```
← → ↻ ⚠ Not secure 184.195.182.212:8080/job/Hiring-App-Multi-Stage/1/console ☆ ⓘ ⋮

Jenkins / Hiring-App-Multi-Stage / #1 🔍 ⚙️

[INFO] Analysis report compressed in 20ms, zip size=34.8 kB
[INFO] Analysis report uploaded in 43ms
[INFO] ANALYSIS SUCCESSFUL, you can find the results at: http://172.31.12.251:9000/dashboard?id=hiring-app
[INFO] Note that you will be able to access the updated dashboard once the server has processed the submitted analysis report
[INFO] More about the report processing at http://172.31.12.251:9000/api/ce/task?id=4c7968b9-555d-4ddf-8f2e-6fdb619308e1
[INFO] Analysis total time: 4.046 s
[INFO] SonarScanner Engine completed successfully
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 7.951 s
[INFO] Finished at: 2026-10-08T10:14:42Z
[INFO] -----
[Pipeline] }
[Pipeline] // withSonarQubeEnv
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Artifact)
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] withEnv
[Pipeline] {
[Pipeline] sh
+ echo 'Verifying WAR'
Verifying WAR
[Pipeline] }
```

```
← → ↻ Not secure 184.195.182.212:8080/job/Hiring-App-Multi-Stage/1/console ☆ ⓘ
Jenkins / Hiring-App-Multi-Stage / #1 🔍 ⚙️

+ echo 'Verifying WAR'
Verifying WAR
+ test -f target/hiring.war
+ ls -lh target/hiring.war
-rw-r--r-- 1 jenkins jenkins 1.7K Oct 8 10:14 target/hiring.war
[Pipeline] nexusArtifactUploader
Uploading artifact hiring.war started....
GroupId: in.javahome
ArtifactId: hiring
Classifier:
Type: war
Version: 0.1
File: hiring.war
Repository:maven_releases
Uploading: http://172.31.47.243:8081/repository/maven_releases/in/javahome/hiring/0.1/hiring-0.1.war
100 % completed (1.7 kB / 1.7 kB).
Uploaded: http://172.31.47.243:8081/repository/maven_releases/in/javahome/hiring/0.1/hiring-0.1.war (1.7 kB at 30 kB/s)
uploading artifact hiring.war completed.
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deployment)
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] withEnv
```

```
← → ↻ Not secure 184.195.182.212:8080/job/Hiring-App-Multi-Stage/1/console ☆ ⓘ
Jenkins / Hiring-App-Multi-Stage / #1 🔍 ⚙️

[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] withEnv
[Pipeline] {
[Pipeline] slackSend
Slack Send Pipeline step running, values are - baseUrl: <empty>, teamDomain: Jenkins, channel: all-jenkins, color: good, botUser: true,
tokenCredentialId: slack-jenkins-token, notifyCommitters: false, iconEmoji: <empty>, username: <empty>, timestamp: <empty>
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
Hiring application deployed successfully.
[Pipeline] echo
http://172.31.33.74:8080/hiring/
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

J Jenkins 🔧 📄

Home

DMs

Activity

Files

Agents & tools

Admin

An offer awaits

Find a conversation...

Huddles

Directories

Starred

Channels

all-jenkins

new-channel

social

Add channels

Direct messages

ravutla.10 you

Invite people

Agents & apps

Jenkins

Describe what you are looking for 🔍

☆ #all-jenkins Invite teammates 🗨 Huddle 🔔 🔍 ⓘ

Messages 📄 Add canvas +

Today ▾

SonarQube: <http://172.31.12.251:9000>

Nexus: <http://172.31.47.243:8081>

Tomcat: <http://172.31.33.74:8080/hiring/>

Jenkins: <http://184.195.128.132:8080/job/Hiring-App-Scripted/3/>

Jenkins AGENT 3:44 PM

Hiring App CI/CD SUCCESS

Application : HiringApp

Build : #1

Branch : main

Tomcat : <http://172.31.33.74:8080/hiring/>

Jenkins : <http://184.195.128.132:8080/job/Hiring-App-Multi-Stage/1/>

B I U ↻ 🔗 📄 📌 📁 📧 📧

Message #all-jenkins

+ ⌨ 🌐 📎 📧 📧 📧 📧

Create one parallel stage pipeline job.

Implementation Steps:

1. Login to **Jenkins**.
2. Go to **Jenkins Dashboard** → **New Item**.
3. Enter the job name:
Hiring-App-Parallel
4. Select **Pipeline**.
5. Click **OK**.
6. Scroll to the **Pipeline** section.
7. Under **Definition**, select:
Pipeline script
8. Enter the complete **Parallel Pipeline** code.

```
pipeline {  
    agent any  
  
    tools {  
        maven 'Maven-3.9.4'  
    }  
  
    environment {  
        APP_NAME = 'HiringApp'  
  
        GIT_URL = 'https://github.com/betawins/hiring-app.git'  
        GIT_BRANCH = 'main'  
  
        SONAR_SERVER = 'SonarQube'  
        SONAR_URL = 'http:// <Private-IP>:9000'
```

NEXUS_URL = '<Private-IP>:8081'

NEXUS_REPOSITORY = 'maven_releases'

NEXUS_CREDENTIALS = 'nexus-admin'

TOMCAT_URL = 'http:// <Private-IP>:8080'

TOMCAT_CREDENTIALS = 'tomcat-jenkins'

TOMCAT_CONTEXT = 'hiring'

SLACK_CHANNEL = 'all-jenkins'

SLACK_CREDENTIALS = 'slack-jenkins-token'

GROUP_ID = 'in.javahome'

ARTIFACT_ID = 'hiring'

APP_VERSION = '0.1'

WAR_FILE = 'target/hiring.war'

}

stages {

stage('Git Clone') {

steps {

deleteDir()

git branch: "\${GIT_BRANCH}", url: "\${GIT_URL}"

sh ""

echo "=====

echo "GIT CLONE"

```

        echo "=====

git log -1 --oneline
'''
    }
}

stage('Maven Build') {
    steps {
        sh '''
            echo "=====

            echo "MAVEN BUILD"

            echo "=====

            mvn -version

            mvn clean package -DskipTests

            ls -lh target/hiring.war
        '''
    }
}

stage('Parallel Checks') {
    parallel {

        stage('SonarQube Analysis') {
            steps {
                withSonarQubeEnv("${SONAR_SERVER}") {
                    sh '''
                        echo "=====

                        echo "SONARQUBE ANALYSIS"
                    '''
                }
            }
        }
    }
}

```

```

        echo "=====

mvn \

org.sonarsource.scanner.maven:sonar-maven-plugin:sonar \

-Dsonar.projectKey=hiring-app \

-Dsonar.projectName=HiringApp \

-Dsonar.sources=. \

-Dsonar.host.url=$SONAR_HOST_URL

'''

    }

}

}

stage('Verify WAR') {
    steps {
        sh '''

        echo "=====

        echo "VERIFY WAR"

        echo "=====

        if [ ! -f "${WAR_FILE}" ]; then

            echo "ERROR: hiring.war was not generated."

            ls -lah target/

            exit 1

        fi

        echo "WAR found:"

        ls -lh "${WAR_FILE}"

```



```

        echo ""

        echo "WAR contents:"

        unzip -l "${WAR_FILE}"
    ""
}
}
}
}

```

```

stage('Nexus Upload') {
    steps {
        nexusArtifactUploader(
            nexusVersion: 'nexus3',
            protocol: 'http',
            nexusUrl: "${NEXUS_URL}",
            groupId: "${GROUP_ID}",
            version: "${APP_VERSION}",
            repository: "${NEXUS_REPOSITORY}",
            credentialsId: "${NEXUS_CREDENTIALS}",
            artifacts: [
                [
                    artifactId: "${ARTIFACT_ID}",
                    classifier: "",
                    file: "${WAR_FILE}",
                    type: 'war'
                ]
            ]
        )
    }
}

```

```
)  
}  
}
```

```
stage('Deploy Tomcat') {  
    steps {  
        deploy(  
            adapters: [  
                tomcat9(  
                    credentialsId: "${TOMCAT_CREDENTIALS}",  
                    url: "${TOMCAT_URL}",  
                    path: "  
                )  
            ],  
            contextPath: "${TOMCAT_CONTEXT}",  
            war: "${WAR_FILE}"  
        )  
    }  
}
```

```
stage('Slack Notification') {  
    steps {  
        slackSend(  
            channel: "${SLACK_CHANNEL}",  
            color: 'good',  
            tokenCredentialId: "${SLACK_CREDENTIALS}",  
            botUser: true,  
            message: ""
```

Hiring App CI/CD SUCCESS

Application : \${APP_NAME}

Build : #\${BUILD_NUMBER}

Branch : \${GIT_BRANCH}

Tomcat : \${TOMCAT_URL}/\${TOMCAT_CONTEXT}/

Jenkins : \${BUILD_URL}

""""

```
    )  
  }  
}  
}
```

```
post {  
  failure {  
    slackSend(  
      channel: "${SLACK_CHANNEL}",  
      color: 'danger',  
      tokenCredentialId: "${SLACK_CREDENTIALS}",  
      botUser: true,  
      message: ""
```

Hiring App CI/CD FAILED

Application : \${APP_NAME}

Build : #\${BUILD_NUMBER}

Please check Jenkins console output.

\${BUILD_URL}

""""

```
    )  
  }  
}
```

```

success {
    echo "Hiring application deployed successfully."
    echo "${TOMCAT_URL}/${TOMCAT_CONTEXT}"/
}
}
}

```

9. Click **Apply**.

10. Click **Save**.

11. Click **Build Now**.

12. Open the generated **Build Number**.

13. Click **Console Output**.

14. Verify the stages execute as:

Git Clone

→ Maven Build

→ Parallel Checks

└─ SonarQube Analysis

└─ Verify WAR

→ Nexus Upload

→ Deploy Tomcat

→ Slack Notification

15. Verify **SonarQube Analysis** completes successfully.

16. Verify **WAR verification** completes successfully.

17. Verify the WAR is uploaded successfully to **Nexus**.

18. Verify the application is successfully deployed to **Tomcat**.

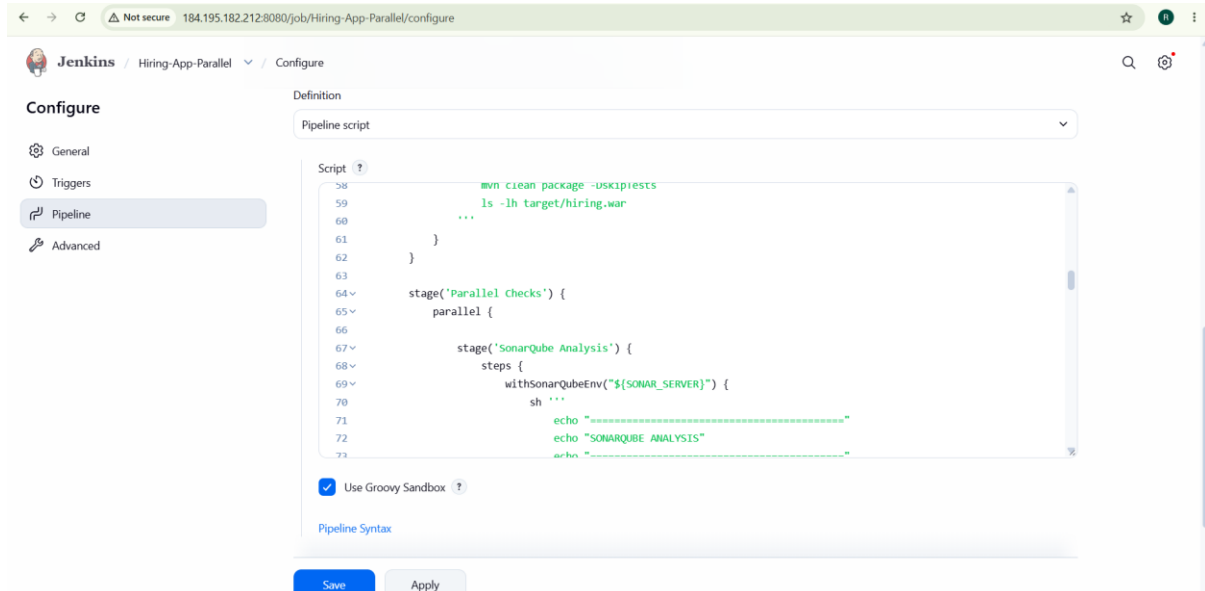
19. Verify the **Slack success notification** is received.

20. Verify the application is accessible:

<http://<Tomcat-Public-IP>:8080/hiring/>

21. Verify the final result:

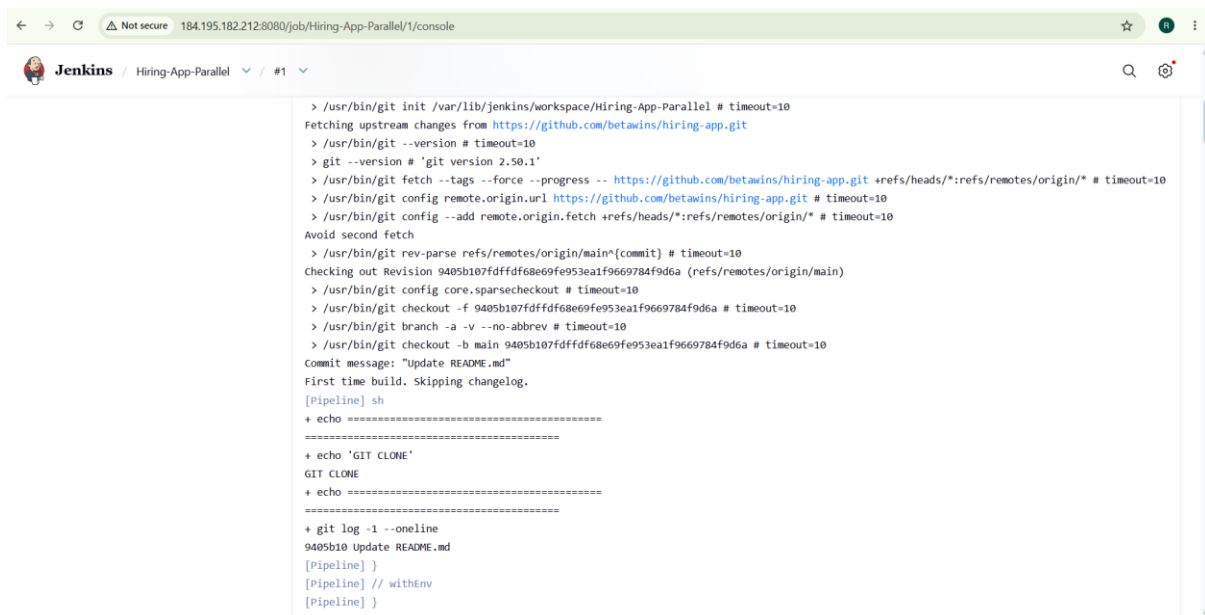
Finished: SUCCESS



The screenshot shows the Jenkins 'Configure' page for a pipeline named 'Hiring-App-Parallel'. The 'Pipeline script' tab is selected, displaying a Groovy script. The script defines a pipeline with a 'Parallel Checks' stage containing a 'SonarQube Analysis' step. The 'Use Groovy Sandbox' checkbox is checked. The 'Save' button is highlighted.

```
58 mvn clean package -DskipTests
59 ls -lh target/hiring.war
60 ...
61 }
62 }
63
64 stage('Parallel Checks') {
65     parallel {
66
67         stage('SonarQube Analysis') {
68             steps {
69                 withSonarQubeEnv("${SONAR_SERVER}") {
70                     sh ...
71                     echo "=====
72                     echo "SONARQUBE ANALYSIS"
73                     echo "=====

```



The screenshot shows the Jenkins console output for the 'Hiring-App-Parallel' pipeline. The output displays the execution of various commands, including git init, git fetch, git checkout, and git log. The pipeline is marked as 'First time build, skipping changelog.' and the final output shows the commit message 'Update README.md'.

```
> /usr/bin/git init /var/lib/jenkins/workspace/Hiring-App-Parallel # timeout=10
Fetching upstream changes from https://github.com/betawins/hiring-app.git
> /usr/bin/git --version # timeout=10
> git --version # 'git version 2.50.1'
> /usr/bin/git fetch --tags --force --progress -- https://github.com/betawins/hiring-app.git +refs/heads/*:refs/remotes/origin/* # timeout=10
> /usr/bin/git config remote.origin.url https://github.com/betawins/hiring-app.git # timeout=10
> /usr/bin/git config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
> /usr/bin/git rev-parse refs/remotes/origin/main^{commit} # timeout=10
Checking out Revision 9405b107fdffdf68e69fe953eaf9669784f9d6a (refs/remotes/origin/main)
> /usr/bin/git config core.sparsecheckout # timeout=10
> /usr/bin/git checkout -f 9405b107fdffdf68e69fe953eaf9669784f9d6a # timeout=10
> /usr/bin/git branch -a -v --no-abbrev # timeout=10
> /usr/bin/git checkout -b main 9405b107fdffdf68e69fe953eaf9669784f9d6a # timeout=10
Commit message: "Update README.md"
First time build, skipping changelog.
[Pipeline] sh
+ echo =====
+ echo 'GIT CLONE'
GIT CLONE
+ echo =====
+ git log -1 --oneline
9405b10 Update README.md
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage

```

```
← → ↻ ⚠ Not secure 184.195.182.212:8080/job/Hiring-App-Parallel/1/console ☆ ⓘ ⋮

Jenkins / Hiring-App-Parallel ▾ / #1 ▾ 🔍 ⚙️

[INFO] skip non existing resourceDirectory /var/lib/jenkins/workspace/Hiring-App-Parallel/src/main/resources
[INFO]
[INFO] --- compiler:3.11.0:compile (default-compile) @ hiring ---
[INFO] No sources to compile
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ hiring ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /var/lib/jenkins/workspace/Hiring-App-Parallel/src/test/resources
[INFO]
[INFO] --- compiler:3.11.0:testCompile (default-testCompile) @ hiring ---
[INFO] No sources to compile
[INFO]
[INFO] --- surefire:3.1.2:test (default-test) @ hiring ---
[INFO] Tests are skipped.
[INFO]
[INFO] --- war:3.3.1:war (default-war) @ hiring ---
[INFO] Packaging webapp
[INFO] Assembling webapp [hiring] in [/var/lib/jenkins/workspace/Hiring-App-Parallel/target/hiring]
[INFO] Processing war project
[INFO] Copying webapp resources [/var/lib/jenkins/workspace/Hiring-App-Parallel/src/main/webapp]
[INFO] Building war: /var/lib/jenkins/workspace/Hiring-App-Parallel/target/hiring.war
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 1.128 s
[INFO] Finished at: 2026-10-08T10:23:18Z
[INFO] -----
+ ls -lh target/hiring.war
-rw-r--r-- 1 jenkins jenkins 1.7K Oct 8 10:23 target/hiring.war
```

```
← → ↻ ⚠ Not secure 184.195.182.212:8080/job/Hiring-App-Parallel/1/console ☆ ⓘ ⋮

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WAR contents:
+ unzip -l target/hiring.war
Archive: target/hiring.war
  Length   Date    Time    Name
  ----
  81  10-08-2026  10:23  META-INF/MANIFEST.MF
  0   10-08-2026  10:23  META-INF/
  0   10-08-2026  10:23  WEB-INF/
  0   10-08-2026  10:23  WEB-INF/classes/
 215  10-08-2026  10:23  WEB-INF/web.xml
  86  10-08-2026  10:23  index.jsp
 890  10-08-2026  10:23  META-INF/maven/in.javahome/hiring/pom.xml
  50  10-08-2026  10:23  META-INF/maven/in.javahome/hiring/pom.properties
-----
1322                                8 files
[Pipeline] }

[SonarQube Analysis] + echo =====
+ echo 'SONARQUBE ANALYSIS'
SONARQUBE ANALYSIS
+ echo =====
+ mvn org.sonarsource.scanner.maven:sonar-maven-plugin:sonar -Dsonar.projectKey=hiring-app -Dsonar.projectName=HiringApp -Dsonar.sources=. -
Dsonar.host.url=http://172.31.12.251:9000/
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
```

```
← → ↻ Not secure 184.195.182.212:8080/job/Hiring-App-Parallel/1/console ☆ ⓘ ⋮

Jenkins / Hiring-App-Parallel ▾ / #1 ▾ 🔍 ⓘ ⋮

[INFO] SCM revision ID '9405b107fdffdf68e69fe953ea1f9669784f9d6a'
[INFO] Analysis report generated in 109ms, dir size=296.2 kB
[INFO] Analysis report compressed in 23ms, zip size=34.9 kB
[INFO] Analysis report uploaded in 32ms
[INFO] ANALYSIS SUCCESSFUL, you can find the results at: http://172.31.12.251:9000/dashboard?id=hiring-app
[INFO] Note that you will be able to access the updated dashboard once the server has processed the submitted analysis report
[INFO] More about the report processing at http://172.31.12.251:9000/api/ce/task?id=148cc81f-b89a-401c-9eba-2db9b6f4240c7
[INFO] Analysis total time: 3.722 s
[INFO] SonarScanner Engine completed successfully
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 7.717 s
[INFO] Finished at: 2026-10-08T10:23:27Z
[INFO] -----
[Pipeline] }
[Pipeline] // withSonarQubeEnv
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // parallel
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Nexus Upload)
[Pipeline] tool
```

```
← → ↻ Not secure 184.195.182.212:8080/job/Hiring-App-Parallel/1/console ☆ ⓘ ⋮

Jenkins / Hiring-App-Parallel ▾ / #1 ▾ 🔍 ⓘ ⋮

[Pipeline] {
[Pipeline] nexusArtifactUploader
Uploading artifact hiring.war started....
GroupId: in.javahome
ArtifactId: hiring
Classifier:
Type: war
Version: 0.1
File: hiring.war
Repository:maven_releases
Uploading: http://172.31.47.243:8081/repository/maven\_releases/in/javahome/hiring/0.1/hiring-0.1.war
100 % completed (1.7 kB / 1.7 kB).
Uploaded: http://172.31.47.243:8081/repository/maven\_releases/in/javahome/hiring/0.1/hiring-0.1.war (1.7 kB at 24 kB/s)
Uploading artifact hiring.war completed.
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy Tomcat)
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] withEnv
[Pipeline] {
[Pipeline] deploy
[DeployPublisher][INFO] Attempting to deploy 1 war file(s)
[DeployPublisher][INFO] Deploying /var/lib/jenkins/workspace/Hiring-App-Parallel/target/hiring.war to container Tomcat 9.x Remote with context hiring
```

